
Password Cracking Project

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Password Cracking

Password cracking is the process of recovering a password from stored data or a protected file. Security professionals use it to test how strong a password is and to show why weak passwords are risky. If a password is short or common, it can be found quickly, which proves the need for strong passwords. Many files like PDF, ZIP, and Office documents can be locked with a password. When a file is locked, its password is stored in the form of a hash. A hash is a scrambled value that represents the password. To recover the password, we first take out this hash from the file, and then run it through a cracking tool that tries different words until it finds a match.

Module 1: Jack the Ripper (JTR) & Johnny

John the Ripper (JTR) is a popular password cracking tool used by security professionals to test password strength. It can check many types of password hashes and also unlock password protected files like PDF, ZIP, and Office documents. **Johnny**, on the other hand, is the graphical version built on John the Ripper, which provides a beginner friendly interface. Both tools are widely used in security testing and learning labs to understand password safety.

In this lab task I used **JTR John** and **JTR Johnny** to recover the password of a protected PDF file using the password hash extracted via the online web tool called **Open Hashcrack**. This exercise

helped me learn how password cracking works and emphasizes the importance of strong passwords for protection.

Situation

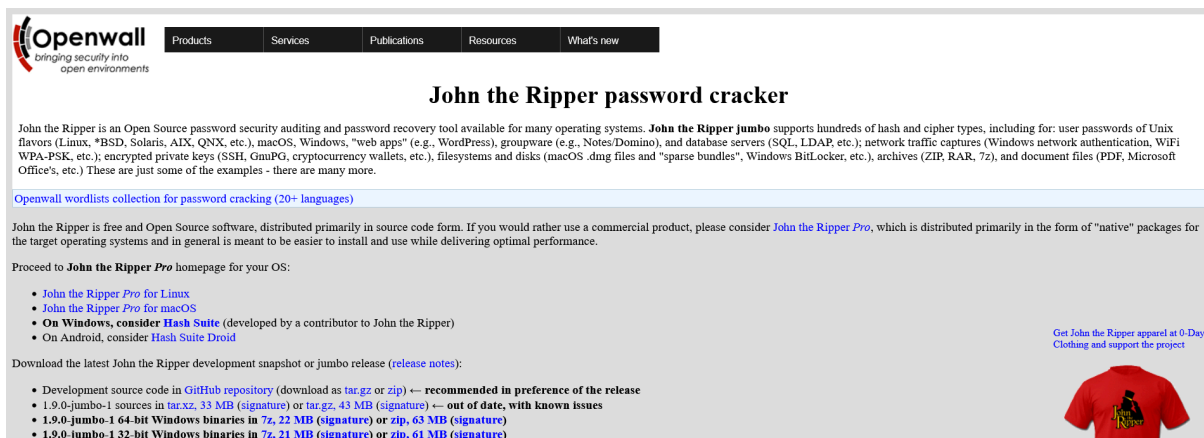
A password-protected PDF file (**My Locked PDF1.pdf**) has been provided but the password was not provided. I am required to crack the password using **Jack the Ripper** and access the file and reveal its contents.

Task

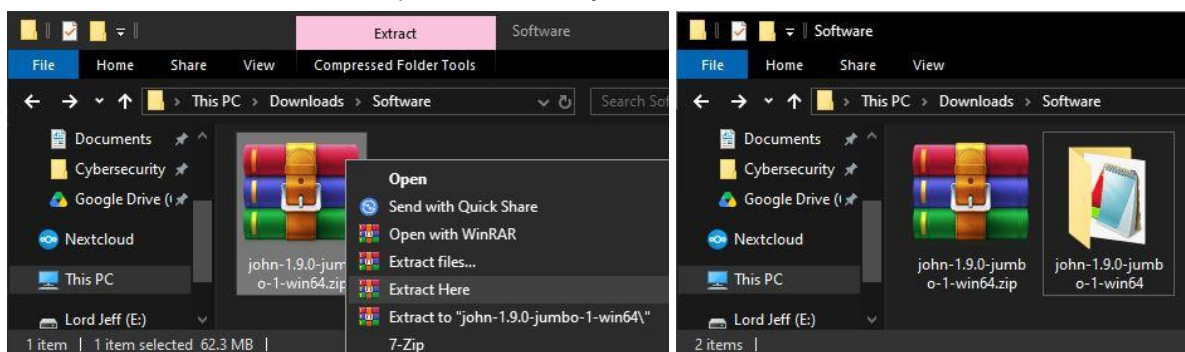
Crack the password of provided PDF file (**My Locked PDF1.pdf**) using the Jack the Ripper (**JTR John & JTR Johnny**) tool on Windows and the online web tool, **Open Hashcrack**.

Action

1. Download the latest Jack the Ripper “jumbo” release.
 - a. Visit the official Jack the Ripper website: <https://www.openwall.com/john>



- b. Download the compatible 64-bit “jumbo” Windows binaries ZIP file.



2. Download the current Johnny release.
 - a. Visit the official Johnny website: <https://openwall.info/wiki/john/johnny>

[[Johnny - GUI for John the Ripper]]

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Johnny - GUI for John the Ripper

Description

Johnny is the cross-platform Open Source GUI frontend for the popular password cracker John the Ripper. It was originally proposed and designed by Shinrok in draft, version 1.0 implementation was achieved by Aleksey Cherepanov as part of GSoC 2012 and Mathieu Laprise took Johnny further towards 2.0 and beyond as part of GSoC 2015.

Johnny's aim is to automate and simplify the password cracking routine with the help of the tremendously versatile and robust John the Ripper, as well as add extra functionality on top of it, specific to Desktop and GUI paradigms, like improved hash and password workflow, multiple attacks and session management, easily define complex attack rules, visual feedback and statistics, all of it on top of the immense capabilities and features offered by both JTR core/proper as well as Jumbo.

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• Binaries 1.1 (OLD)
• Sources
• Current state

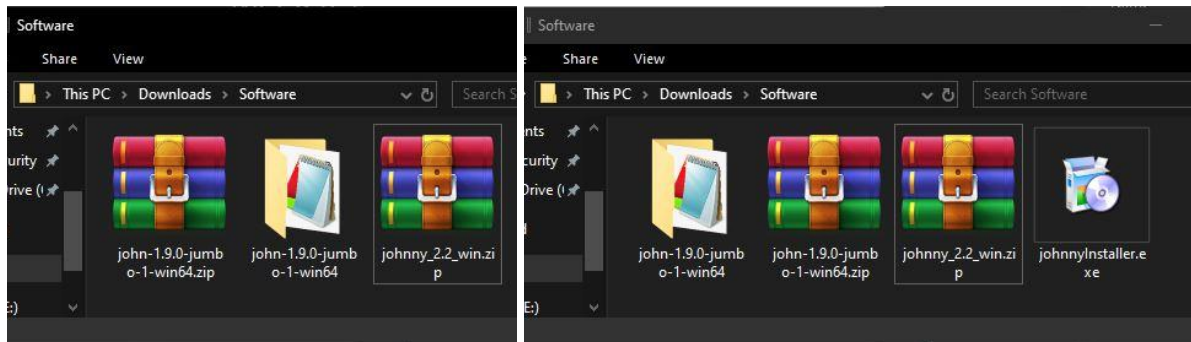
Binary redistributables

The current version is 2.2.

Binaries 2.2 (CURRENT)

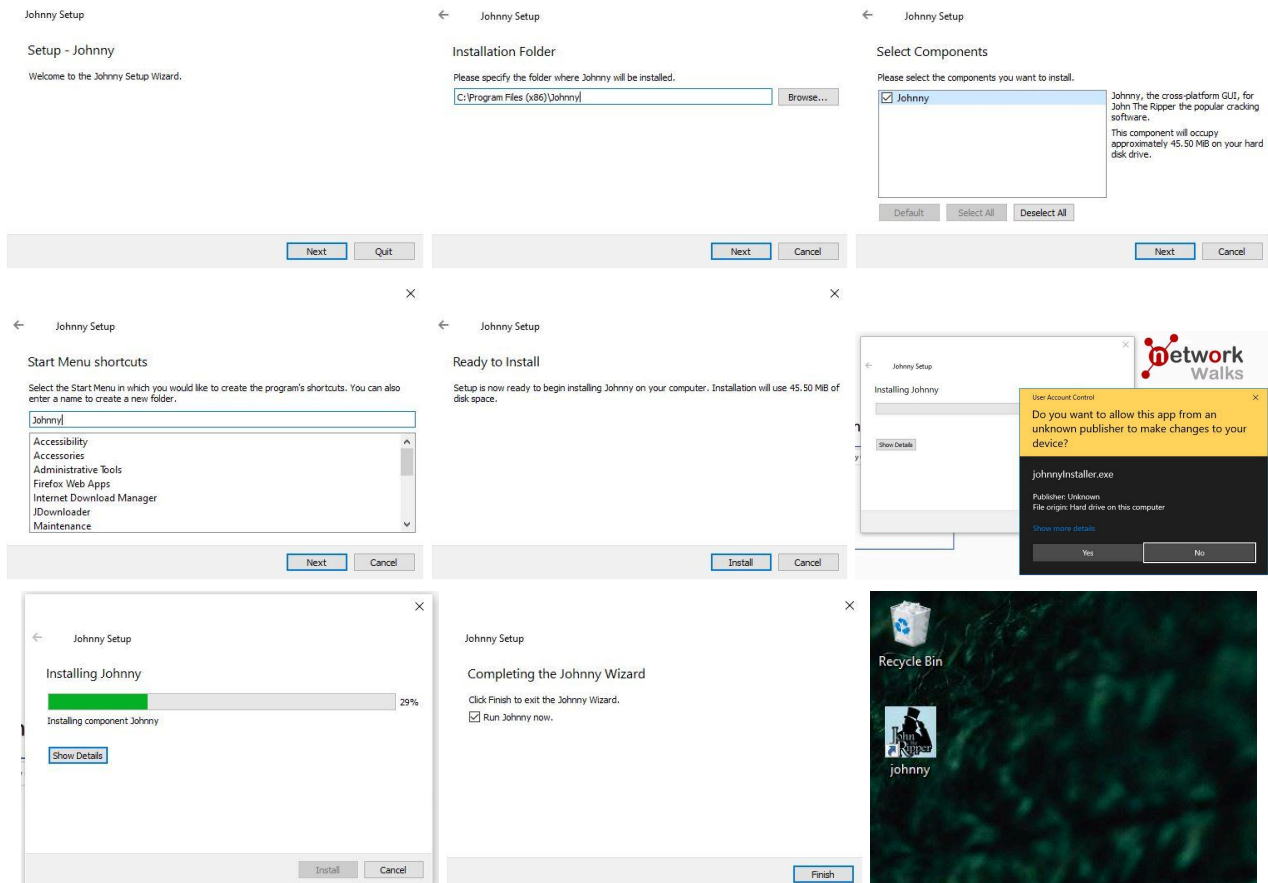
- Windows : [Johnny_2.2_win.zip](#)
- OS X : [Johnny_2.2_osx.zip](#)
- Linux : binaries are not provided, you should build the software from source (see <https://github.com/openwall/Johnny/blob/v2.2/INSTALL> and Sources section).

b. Download the current Windows binaries version.

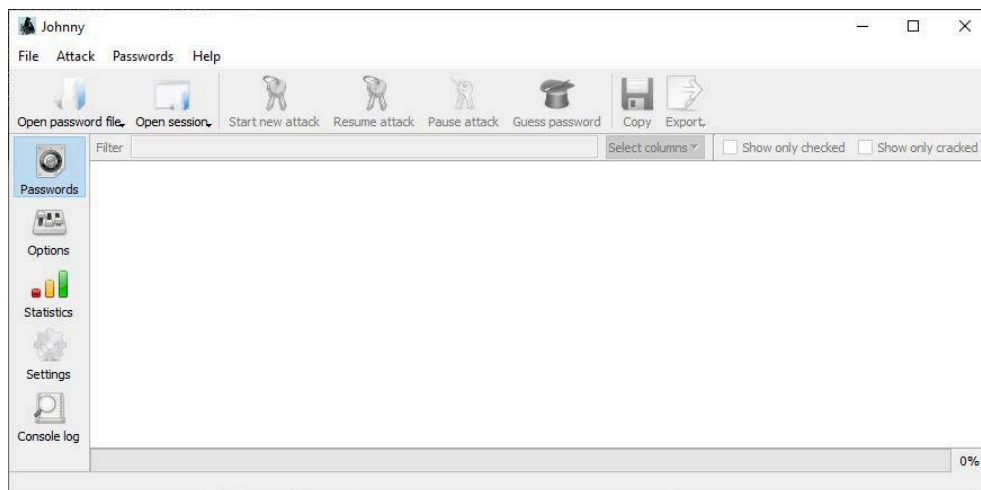


3. Install Johnny

a. Run the setup file and complete the installation wizard.

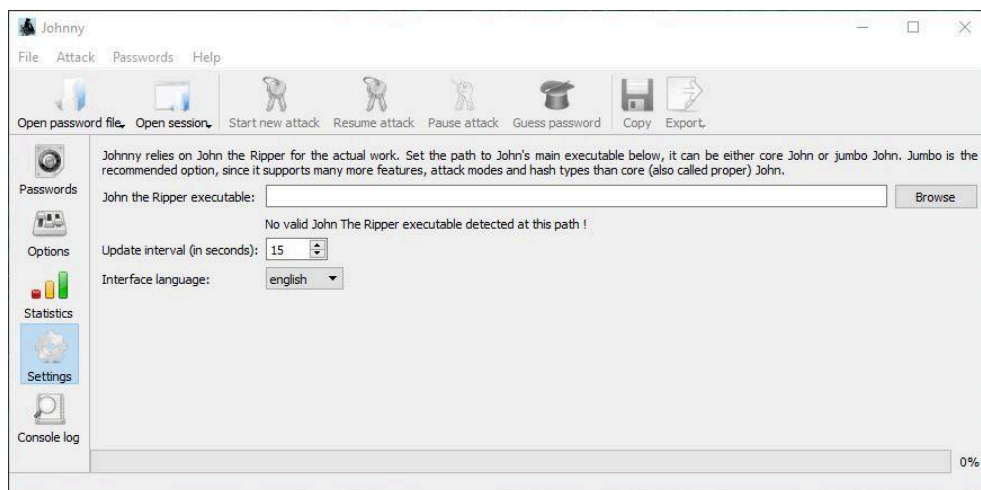


4. Open Johnny.

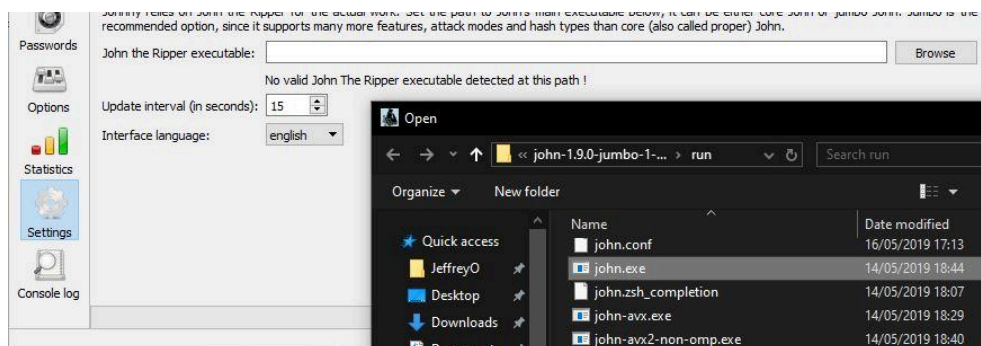


5. Add John.exe to Johnny.

- a. Click **Settings** from the side menu.



- b. Click **Browse**.
- c. Locate **John.exe** and select it



6. Extract the file hash.

- a. Open the hash website:
<https://www.onlinehashcrack.com/tools-pdf-hash-extractor.php>



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PDF HASH EXTRACTOR - INSTANTLY

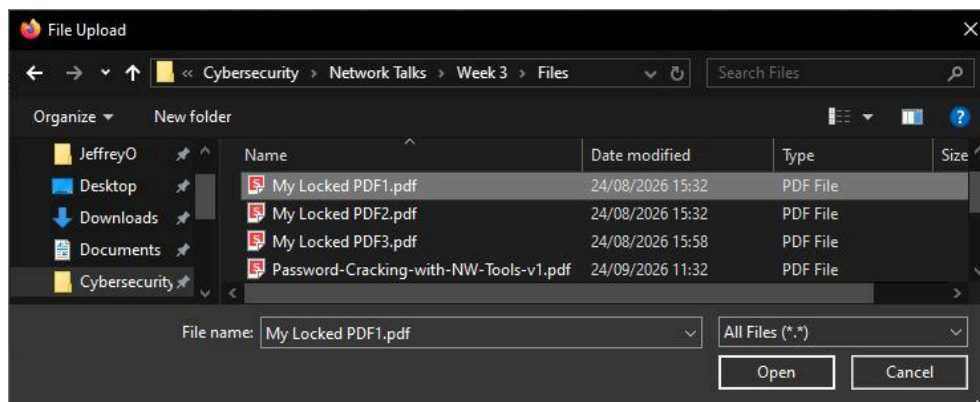
This page will extract the information needed from your PDF to convert it to hash, also known as pdf2john or pdf2hashcat.

This conversion tool is provided for authorized security auditing purposes only. Need full password audit? [Submit](#) your hash securely on our platform.

Select and Upload your PDF:

Select file... [Browse...](#)

b. Click **Browse** and locate the PDF file (**My Locked PDF1.pdf**).



This conversion tool is provided for authorized security auditing purposes only. Need full password audit? [Submit](#) your hash securely on our platform.

Select and Upload your PDF:

My Locked PDF1.pdf [Remove](#) [Upload](#) [Browse...](#)

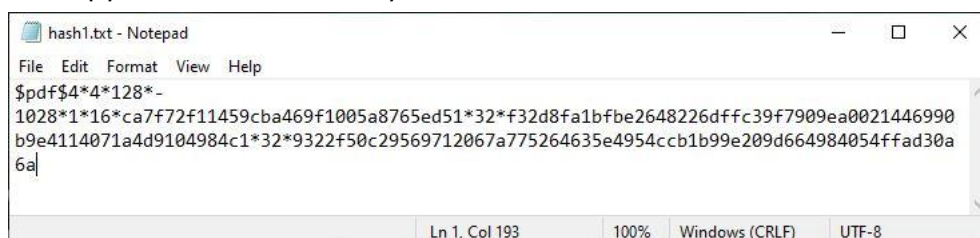
c. Click **Upload**. The hash will appear in the **Output** box.

Select file... [Remove](#) [Upload](#) [Browse...](#)

Output:

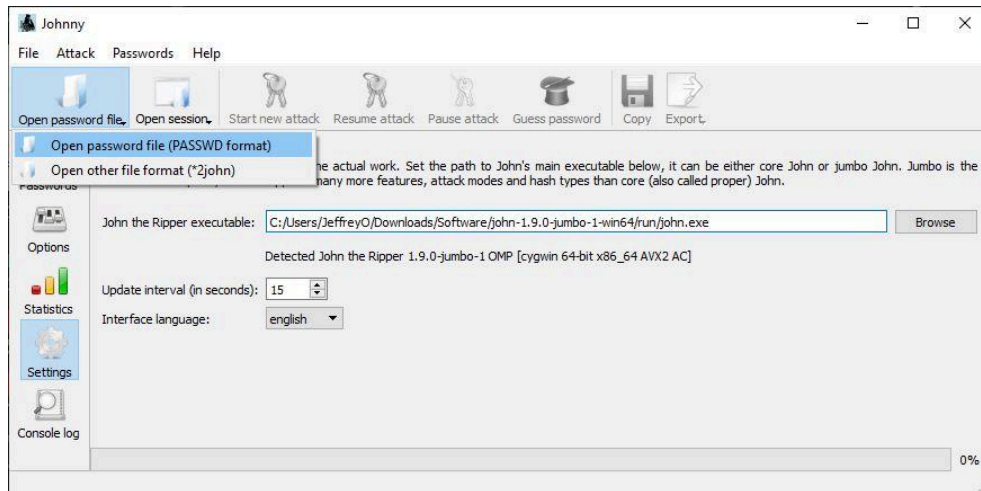
\$pdf\$4*4*128*-1028*1*16*ca7f72f11459cba469f1005a8765ed51*32*f32d8fa1bfbe2648226dfc39f7909ea0021446990b9e4114071a4d9104984c1*32*9322f50c29569712067a775264635e4954ccb1b99e209d664984054ffad30a6a

d. Copy the hash into a newly created TXT file called **“hash1.txt”** and save it.

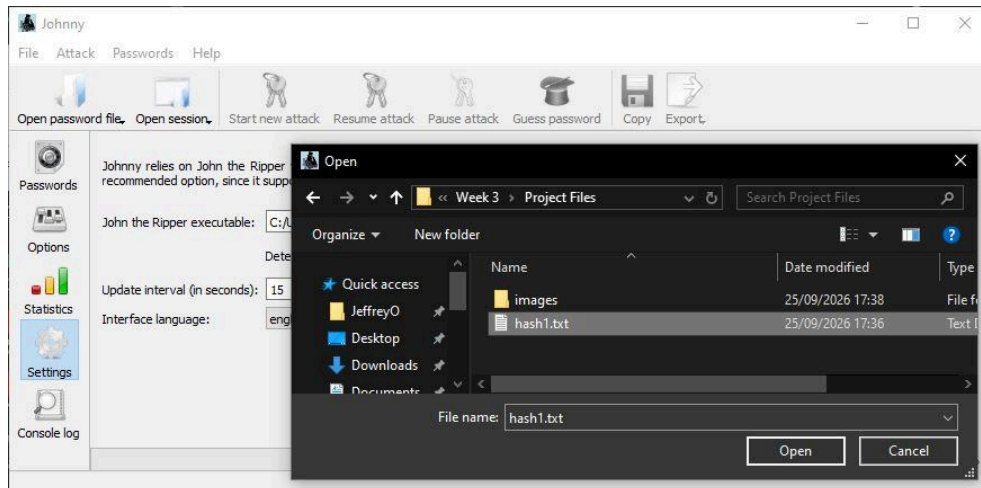


7. Crack the Password.

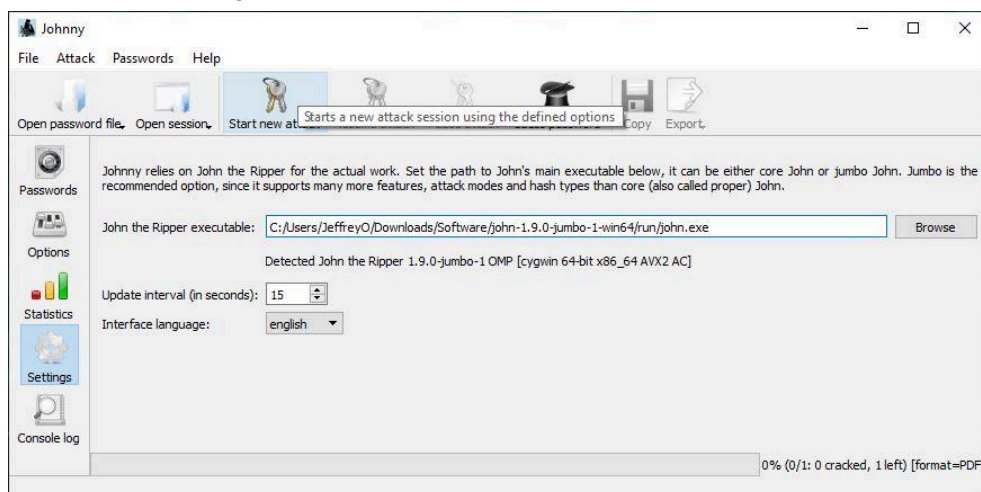
- a. Return to **Johnny** and in the main menu, select **Open password file → Open password file (PASSWD format)**.

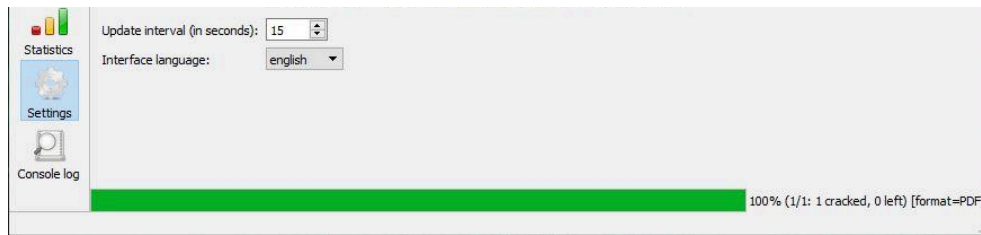


- b. Locate and select the **hash1.txt** file.



- c. On the **Johnny** interface, select **Start new attack**.

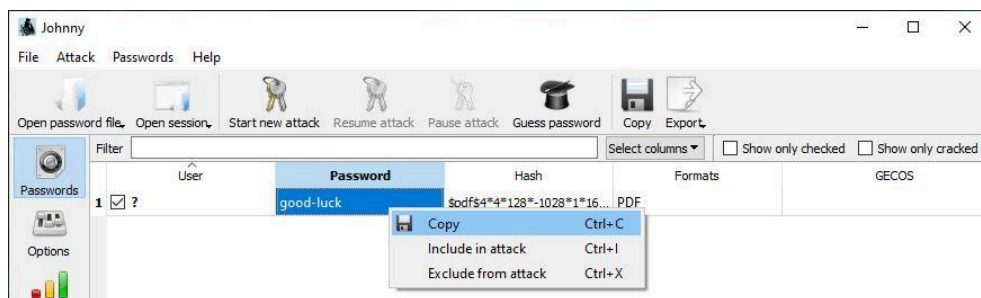




- d. Select **Passwords** from the side menu to open the **Passwords** tab.

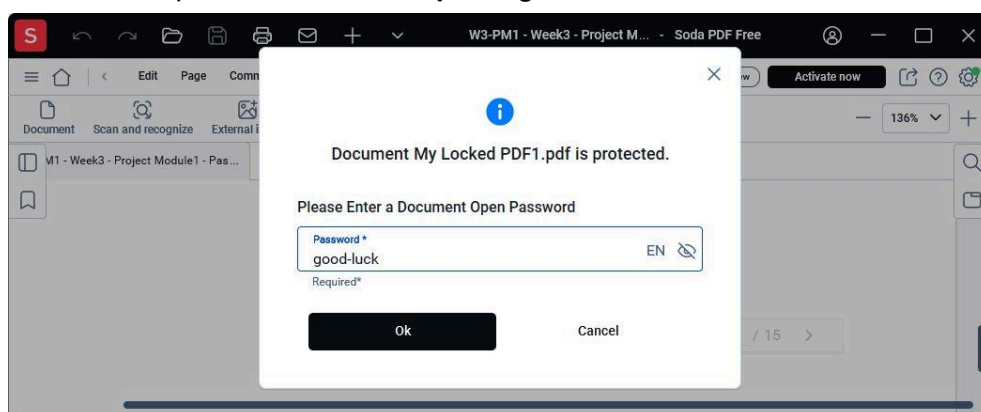


- e. Under the “Password” column (next to the hash from **Step 6c**), is the cracked password: **good-luck**. Copy it.

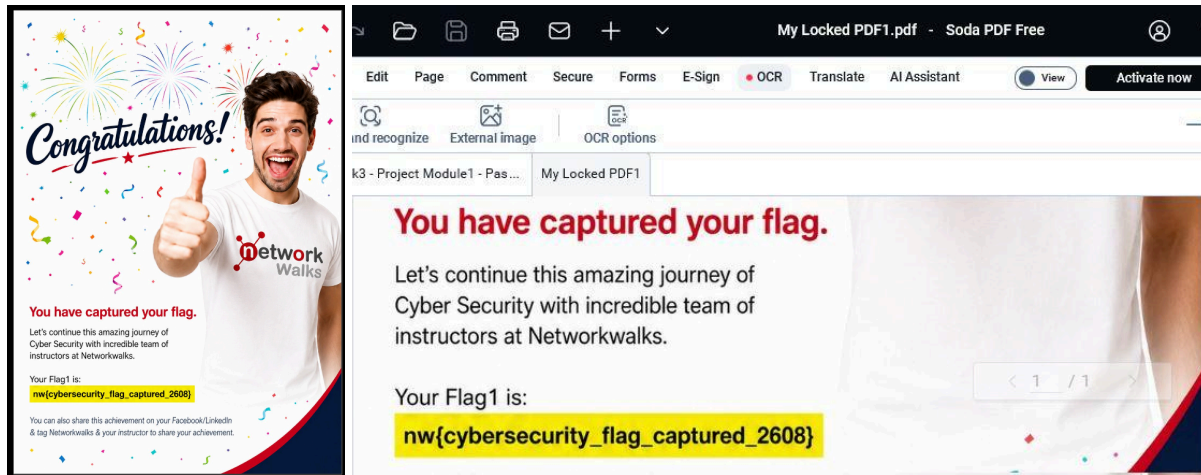


8. Open the PDF file (**My Locked PDF1.pdf**).

- a. Locate the PDF file (**My Locked PDF1.pdf**) and open it.
b. Enter the password (from **Step 7e**): **good-luck**



- c. The file opens with the contents visible.
d. Locate “**Flag1**”: **nw{cybersecurity_flag_captured_2608}**



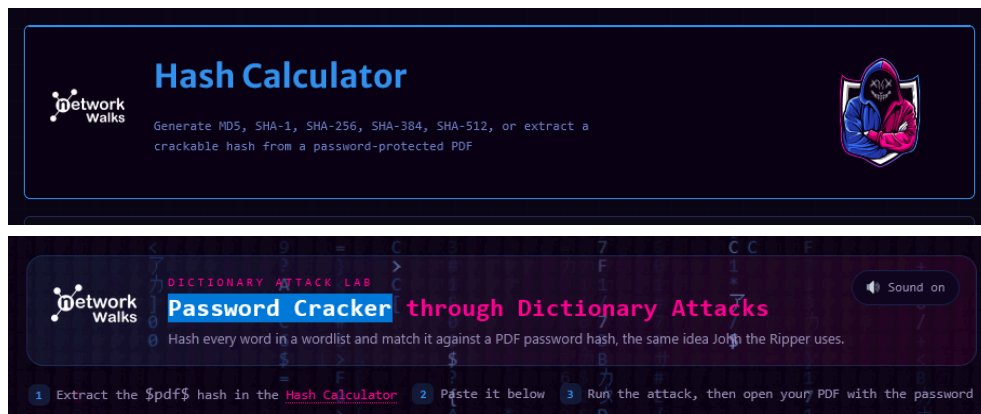
Result

The task is now complete. The flag — `nw{cybersecurity_flag_captured_2608}` — has been retrieved using the required hash extraction and cracking tools.

Lessons Learned

With how fast the password was cracked, I learned the importance of complex passwords. I also learned about the process of file password cracking, from observing the required procedure.

Module 2: Network Walks (NW) Tools



Hash Calculator and Password Cracker are two free web-based security tools created by Network Walks. Hash Calculator is used to take the password hash out of a locked PDF file and Password Cracker is used to find the real password using that hash.

For this project, using the web browser, I implemented both online tools to take the password hash out of the provided locked PDF file (**My Locked PDF2.pdf**) and to find the real password using hash. Both tools ran in the web browser. This lab helped me understand how password cracking works, step by step, and reinforced the importance of strong passwords for file protection.

Situation

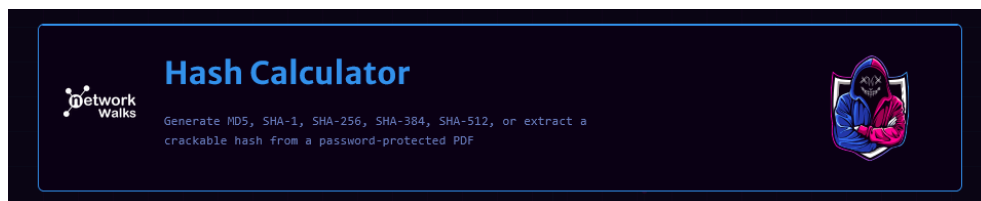
A password-protected PDF file (**My Locked PDF2.pdf**) has been provided but the password was not provided. I am required to crack its password using Network Walks' security tools to access the file and reveal its contents.

Task

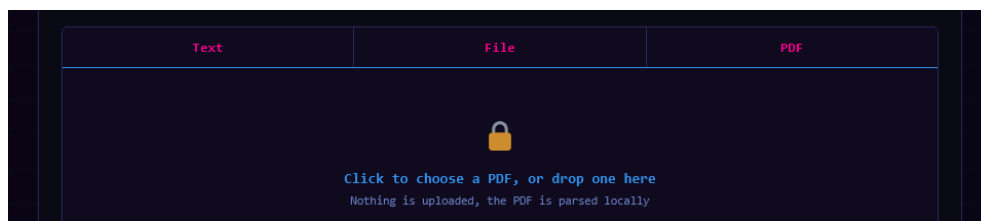
Crack the password of the provided PDF file (**My Locked PDF2.pdf**) using Network Walks' proprietary web tools: **Hash Calculator** and **Password Cracker**, both accessed from the browser.

Action

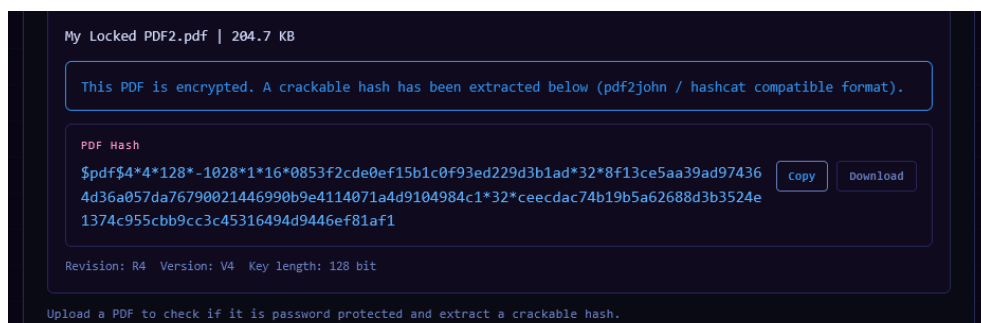
1. Extract the hash using **Hash Calculator**.
 - a. Open Network Walks' **Hash Calculator** in the web browser using the URL:
<https://networkwalks.com/hash-calculator/>

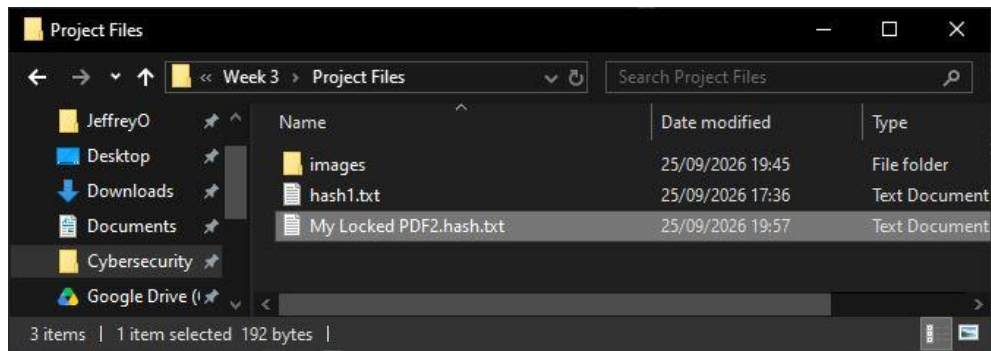


- b. Select the **PDF** tab and click the padlock icon to upload a file. Select the provided PDF file (**My Locked PDF2.pdf**).



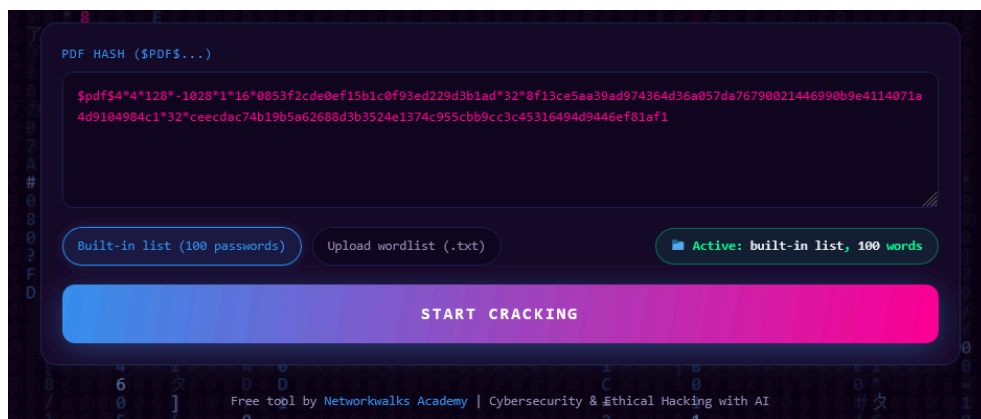
- c. A **PDF Hash** section emerges with the extracted password hash inside it. Click the **Copy** button to copy the hash, then click the **Download** button to download the hash in a TXT file.



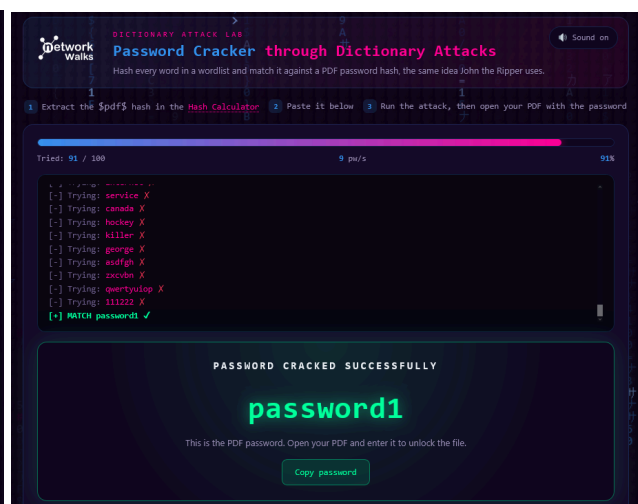
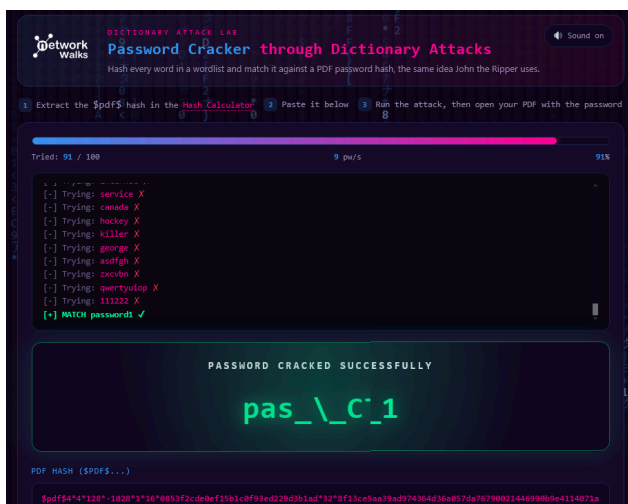


2. Crack the password with **Password Cracker**.

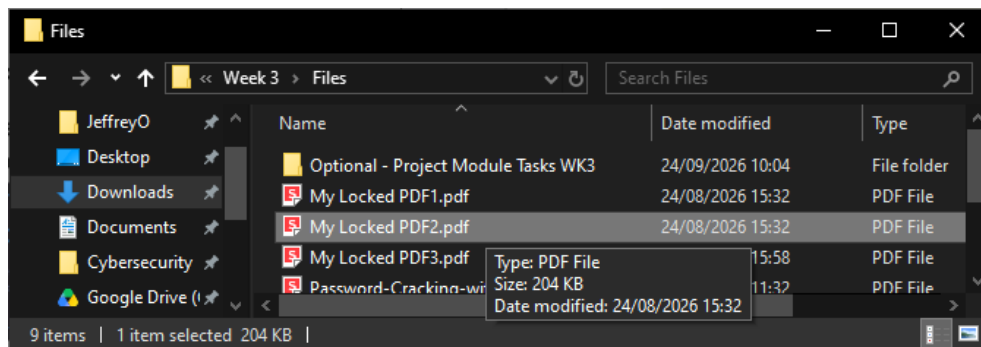
- Via the browser, access the **Password Cracker** interface through the URL:
<https://networkwalks.com/password-cracker/>
- Paste the previously copied hash into the **PDF HASH** box and click the **START CRACKING** button.



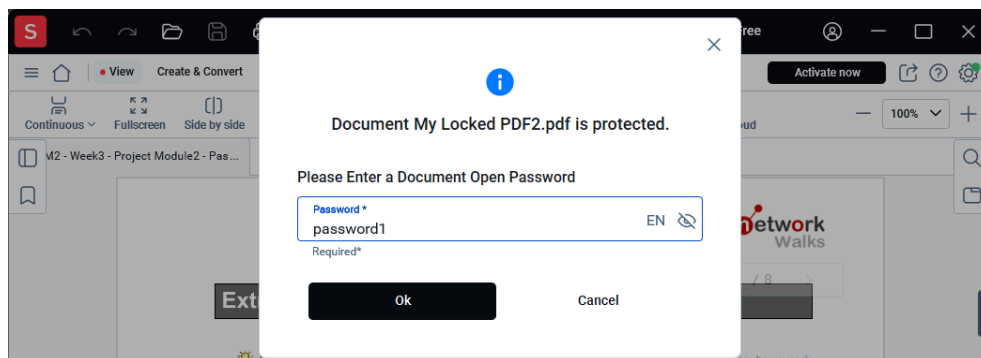
- After the progress bar reaches the end, the extracted password (**password1**) is revealed in the emerged box. Click the **Copy** button to copy the newly revealed password.



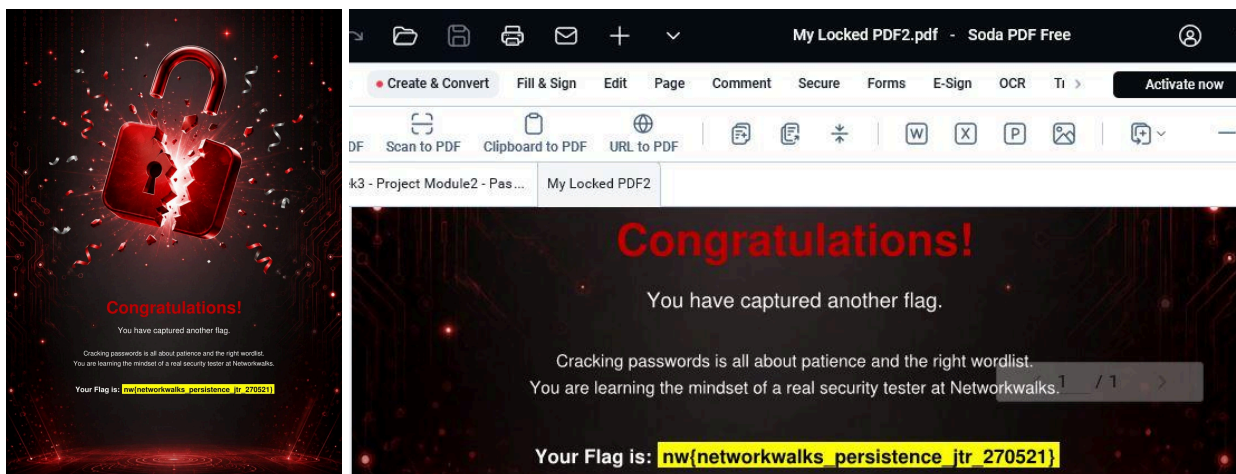
3. Access the encrypted PDF file (**My Locked PDF2.pdf**).
 - a. Locate the password-protected PDF file (**My Locked PDF2.pdf**) and open it.



- b. Enter the password (from **Step 2c**): **password1**



- c. The file opens with the contents visible.
 - d. Locate the flag: **nw{networkwalks_persistence_jtr_270521}**



Result

The task is now complete. The flag — **nw{networkwalks_persistence_jtr_270521}** — has been retrieved using the required hash extraction and cracking tools.

Lessons Learned

Just like the previous attempt, the password was cracked fairly quickly, reinforcing the importance of complex passwords. I also familiarized myself with another pair of tools while implementing the already understood procedure of file password cracking.